Fantine Mpacko Priso

Berkeley, CA / Paris, France | +33 7 82 93 41 27 | fantine.mpackopriso@berkeley.edu | github.com/Fantiflex

Profile

Computer Science and Applied Mathematics researcher. Interested in developing robust and equitable machine learning methods for computational health and human-centered systems.

Education

University of California, Berkeley 2025–2026
Visiting Graduate Student - College of Computing, Data Science & Society

- Relevant coursework : Machine Learning, Deep Neural Networks Architectures & Optimization, Stochastic Processes, Computer Vision, Functional Analysis, Computational precision health & neuronal function.

Paris-Saclay University, ENS Paris-Saclay, France 2021–2026
MSc in Computer Science (on leave)
Double Major BSc in Mathematics and EECS (grad. June 2024)

- Relevant coursework : Reinforcement Learning, Signal processing, information transmission, probability theory, digital electronics, signal processing, linear algebra, real analysis.

Research Experience

Research intern - Cognitive & Computational Neuroscience Summer 2026
LNC2, Ecole Normale Superieure, Paris
Designed and deployed a three-condition behavioral experiment investigating racial minority proportion overestimation in the French context, testing whether the effect reflects social categorization or general perceptual biases. | GitHub

- Integrated controlled visual stimuli, randomized trials, reaction-time measurements, and structured data collection through JATOS.
- Implemented participant-specific face reconstruction using Bradley-Terry preference modeling, MediaPipe landmarks, geometric alignment, and Delaunay-based morphing.
- Built a reproducible Python stimulus pipeline, auditing 597 Chicago Face Database records and selecting an age-matched set of 122 faces to generate 45 paired face–circle matrices.

Selected Projects

Modular Manifolds in Practice: Accelerating MuOn Optimizer with Quasi-Newton Methods | GitHub

- Evaluated Riemannian L-BFGS as a faster inner solver for MuOn across MLPs, CNNs, and Vision Transformers on CIFAR-10/100, demonstrating a 4.4× runtime reduction on CIFAR-10 MLPs with a modest 4–6 percentage-point accuracy tradeoff.

UFO.recordz - Full-Stack Web Development & Data Integration | GitHub | UFOrecordz.fr

- Designed and deployed a responsive web platform using React, TypeScript, Tailwind CSS, and Vercel, integrating Supabase (PostgreSQL) for centralized artist, event, and release management.
- Developed an automated SoundCloud synchronization pipeline using OAuth and Supabase Edge Functions, retrieving and storing 100+ music releases.

EEG-Guided Visual Stimulation System

- Analyzed 16-channel EEG recordings sampled at 500 Hz to characterize occipital brain activity and identify alpha-band oscillations around 11 Hz.
- Implemented a third-order Butterworth filtering and signal-quantization pipeline in Python to translate EEG dynamics into discrete stimulation patterns.

Technical Skills

Programming: Python, C++, MATLAB, Java, SQL, R
Machine Learning: PyTorch, scikit-learn, CUDA
Scientific Computing & Data: NumPy, SciPy, Pandas, Polars
Computer Vision: OpenCV, MediaPipe, Pillow
Research & Development Tools: Git, JATOS, Streamlit, Google Apps Script

Leadership & Community Engagement

Head Student Representative - ENS Paris-Saclay 2024–2025
Represented ∼1,000 students and organized initiatives/events with budgets up to €100K.

Co-Founder & Technical Lead, UFO.recordz 2021–Present
Led digital infrastructure and event operations for a cultural collective reaching up to 2,500 attendees and 500K monthly social-media views.